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Towards last-mile connectivity in 6G-IoT: An integrated MIMO-FSO communication system and FBG sensors under atmospheric attenuations, strain and temperature effects

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ABSTRACT

Free-space optical (FSO) communication is an advanced wireless optical communication technology that provides high-speed data services for 6th-generation wireless communication and Internet of Things (IoT) networks. In light of developing a 6G IoT network in an open environment, this paper analyses the impact of temperature and strain on FSO communication links using Fiber Bragg Grating (FBG) sensors. FSO systems are sensitive to temperature variations, and the position of the receiving telescope can be altered by mechanical strain that can affect the refractive index of the atmosphere, leading to signal attenuation and pointing errors. The current work proposes a model that integrates FBG sensors with an FSO channel for simultaneous strain and temperature measurements and a compensator that overcomes severe signal attenuations. Also, the impact of atmospheric attenuations on FSO systems is analysed. Scintillation models for weak, moderate and strong turbulence conditions are analysed, and the performance of the gamma-gamma turbulence model has been used to observe the MIMO FSO channel. The results are observed for FSO links with the atmospheric attenuations with FBG sensors that reflect the pointing errors at the receiver, showing a comprehensive ability to capture strain and temperature parameters. For the transmission of 10 Gb/s data, it was observed that the inclusion of the Multiple input and multiple-output (MIMO)-FSO technique significantly reduces bit errors from -3.12494 dB to -35.018 dB and increase signal power from 52.4 dBm to 58.9 dBm, indicating the adaptability of this integrated strategy for FSO communication with FBG sensors for last mile connectivity in 6G-IoT applications.

1. Introduction

Development in the Internet of Things (IoT) technology requires high-speed wireless connections and sensor networks. Free space optical (FSO) is an optical wireless communication that supports high-data-rate transmission and offers secure connectivity in IoT networks. Since the FSO link uses point-to-point optical networks and operates in the electromagnetically free optical band, it is suitable for deployment in areas with significant RF signal traffic [1,2]. FBG sensors have also been increasingly integrated into the IoT because of their corrosion resistance, resistance to electromagnetic and radio frequency interference, and small size and weight. The sensing capability of FBGs can be seen in accelerometers for smart healthcare in consumer electronics and environmental monitoring, making them crucial in sensor technology and communications [3,4].

Although FSO stands out as a promising technology to transmit data through free space, offering high bandwidth and low latency, the reliability and performance of FSO systems can be influenced by environmental factors, such as temperature variations, mechanical strains, and atmospheric conditions [5].

Research on FSO communication is essential for advancing the optical communication field and addressing various challenges associated with this technology [6–8]. Addressing these challenges requires innovative solutions, and one such solution lies in integrating FBG sensors with the receiver of the FSO communication system, which are known for their sensitivity to various physical parameters and have demonstrated their efficiency in monitoring environmental effects in diverse applications [9,10]. FSO technology enables the rapid transmission of data between satellites located in space along with 6G IoT applications that can be enhanced via mmWave, FSO, and underwater wireless optical communication [11–14]. These technologies can enhance spectral

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Nomenclature			
Symbol	Description		
P_e	Photoelastic constant	P_r	Received optical power
ϵ	Induced strain	P_t	Transmitted power
ζ	Thermal optic coefficient	D_r	Receiving antenna diameter
ΔT	Temperature variation	D_t	Transmitting antenna diameter
μ_X	Mean of the received intensity	θ	Beam divergence angle
σ_X	Variance	α	Specific attenuation
σ_R^2	Rytov variance that is less than 1 for weak turbulence	Z	Transmission range.
C_n^2	Refractive index structure constant	λ_B	Bragg wavelength
k	Optical wavenumber ($2\pi/\lambda$)	n_e	Effective index
L	FSO link length	Λ	Grating periodicity
σ	Attenuation coefficient	ϵ	Longitudinal strain
V	Visibility	ΔT	Temperature change
λ	Wavelength in nanometres	p_e	Photo-elastic coefficient
$q(v)$	Particle size distribution coefficient that is described using the Kruse model and which is widely used	α_{TE}	Thermal expansion coefficient
		α_{TO}	Thermos optic coefficient
		F	Force by pointing error
		E	Young's modulus
		A	Cross sectional area

Table 1
System properties.

Components/Parameters	Value
PRBS – Bit-rate, Sequence-length	10Gb/s
CW Laser – Wavelength, Transmission power, Linewidth	1552nm, 20dBm, 10 MHz
MZM Extinction ratio	30 dB
Optical amplifiers Gain, Noise figure	20dB, 4dB
FSO links – Transmission telescope aperture, Receiver telescope aperture, Beam Divergence, Index refraction structure	5cm, 20cm, 0.25 mrad $5 \times 10^{-15} m^{-2/3}$
White light source – Frequency, Average Power	1550 nm, –130 dB
FBG Sensor –Wavelength shift for temperature and strain	1545 nm, 1550 nm
PIN Photo-detector's responsivity, dark-current	1A/W, 10nA

Table 2
Visibility and atmospheric losses for 1550 nm wavelength.

Weather condition	Visibility (km)	Attenuation (dB/km)
Thick fog	0.2	75
Moderate fog	0.5	28.9
Light fog/Storm	0.77	18.3
Very light fog/Heavy rain	1.9	6.9
Very light mist/light rain	4	3.1
	5.9	2
Clear air	18.1	0.6
	20	0.54

efficiency, enabling the transmission of large volumes of data within a certain bandwidth.

This work addresses severe signal losses while transmitting user data through the optical wireless communication network. It proposes a novel technique that integrates a MIMO FSO link with an FBG sensor and

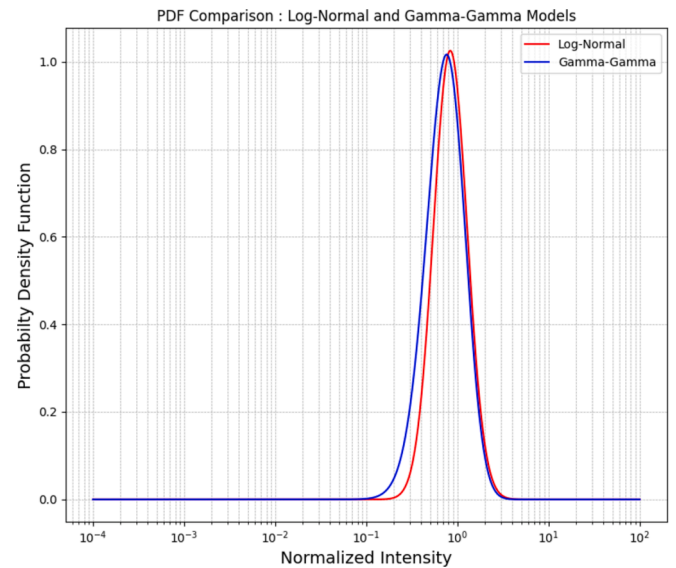


Fig. 2. Probability Density Function comparison of scintillation models.

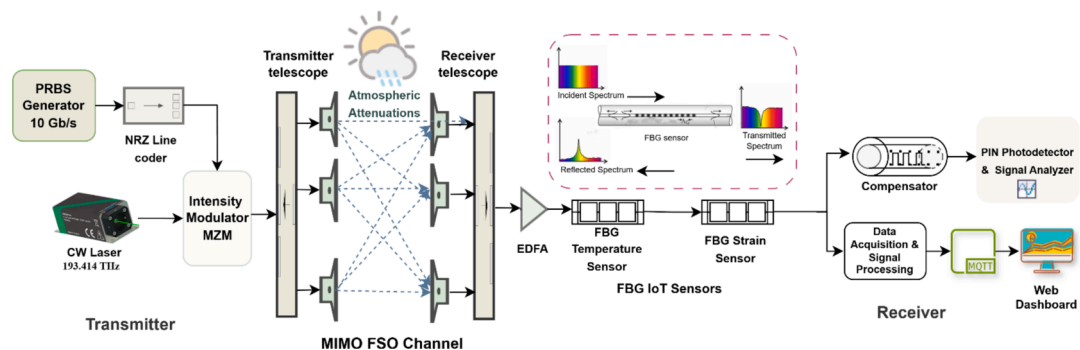


Fig. 1. Proposed integrated FSO communication and FBG sensing system.

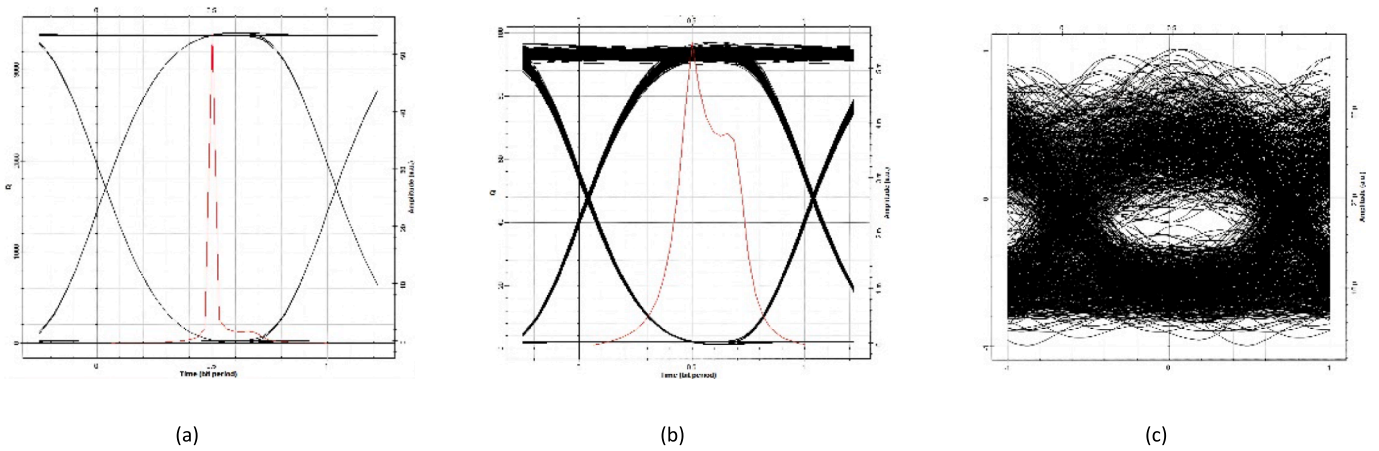


Fig. 3. Eye Diagrams – FSO link range under clear air with attenuation of 0.54 dB/km condition (a) 9.6 km, (b) 28 km, (c) 69.6 km.

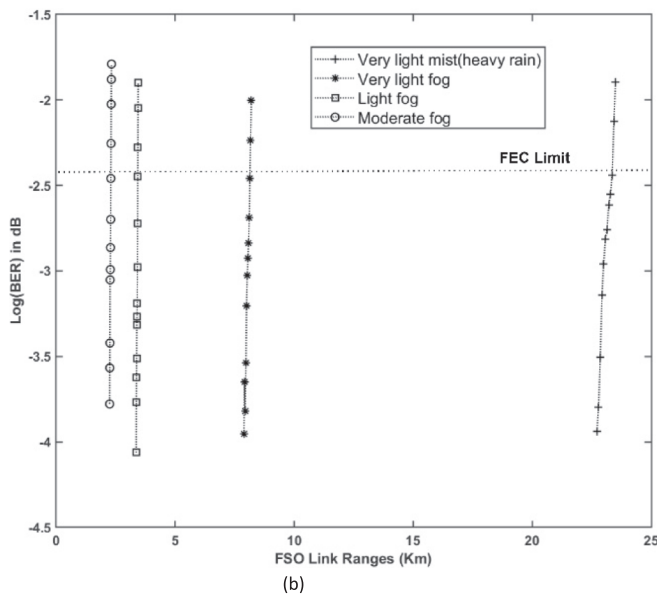
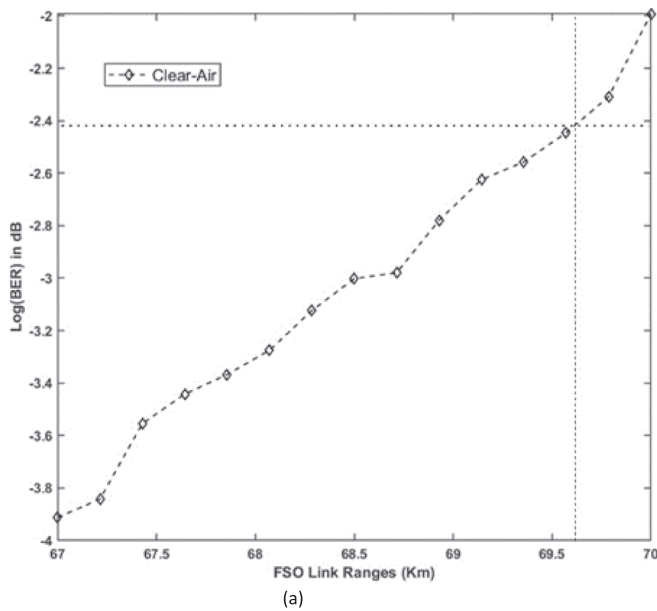


Fig. 4. BER vs FSO link ranges under (a) Clear air – attenuation, (b) very light mist (heavy rain), very light fog, light fog and moderate fog.

compensation at the receiver. Atmospheric turbulence models for weak, moderate, and strong levels are analysed and compared. Then, the system is integrated with FBG sensors that show the impact of misalignment at the receiver. Overall, the purpose of integration is to set up real-time monitoring and quick reactions to structural changes.

Section II reviews the related works, and Section III provides extensive details of system architecture and the causes of signal attenuations. Section IV discusses the results in detail, and Section V concludes the work.

2. Related works

The section provides an overview of several research studies conducted to examine the performance of FSO networks for 6G and IoT applications. Previous research reported that the hybrid FSO system with radio frequency and FSO with visible light communication link in an open environment communication for signal attenuations in different atmospheric conditions, scintillations, and turbulence [15–17]. An adaptive combining technique was proposed by the article [18], in which the authors combined FSO and RF channels. To check the reliability of the system, the closed-form expressions were derived for outage probability and symbol error rate, where authors used the Málaga channel model for analysis and extended their work to a UAV-assisted system in which the Gamma Gamma turbulence model was used [19]. The log-normal distribution model is commonly employed to analyse the FSO networks under weak turbulence. However, the Gamma Gamma model is preferred when addressing moderate to heavy turbulence [20,21]. The article [22] presents the Málaga channel model with a low-density parity check-based MIMO-OFDM-based FSO link. The MIMO analysis becomes complicated because the mathematical analysis uses the Meijer-G function.

In the article [23], the authors proposed a hybrid optical fibre and FSO channel, including an FBG sensor head for monitoring bridge conditions. In case of bridge damage, the transmission path switches from a fibre optic channel to a free-space optical. FBG sensors monitor the physical state of resources and infrastructure in IoT applications. Authors in the article [3] proposed FBG-based accelerometers to monitor the activity levels of elders living alone. In the work [24], the authors tested an FBG sensor arrangement with two Gaussian apodised sensors using OptiGrating. Then, the customised FBGs were put into OptiSystem for analysis. Two FBGs with varying centre wavelengths were chosen to differentiate strain and temperature to test the modelling tools' capacity to deliver and integrate sensor installations. The temperature sensitivity and strain measurements are 1.2 pm/ μ strain and 14.4 pm/ $^{\circ}$ C, respectively. In their work [25], authors compare methods for reducing dispersion in single-mode fibres at 10 Gbps and 40 Gbps by combining uniform FBG and Gaussian-apodized chirped FBG. The simulations

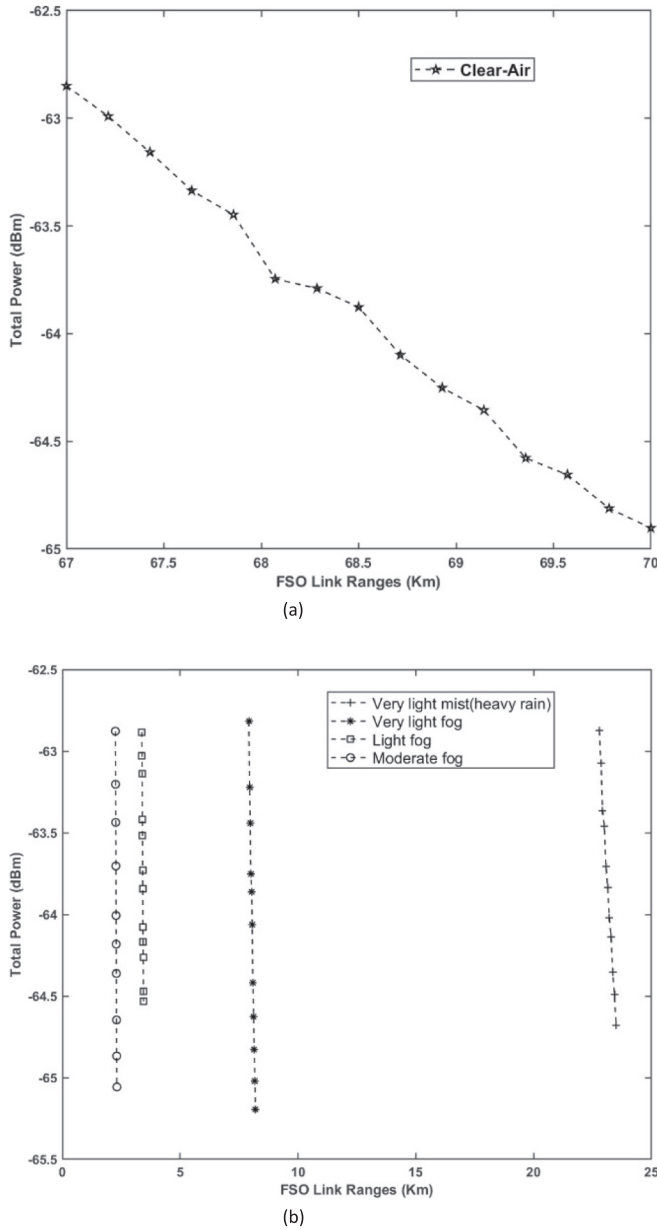


Fig. 5. Received power vs FSO link ranges under (a) Clear air – attenuation, (b) very light mist (heavy rain), very light fog, light fog and moderate fog.

demonstrated that cascading FBG designs increase single-mode fibre reach and quality factors.

Several researchers have analysed research on MIMO-based and wavelength division multiplexed FSO links with various modulation techniques in adverse weather conditions in previous works [26,27]. Authors in [28], developed a methodology to improve bandwidth efficiency for FSO systems using multilevel modulation, OAM, and PPM. Adaptive spatial modulation (SM), spatial pulse position modulation (SPPM), and diversity addressed atmospheric attenuations and optical nonlinearity. In [29], authors showed a hybrid model combining M-ary PPM and SM for MIMO to improve BER, SNR, and sensitivity under various turbulences. The authors in [30] and [31] proposed a spatial multiplexing technique that uses MIMO transceivers to enhance the transmission range of an FSO system. All works investigated the impact of pointing errors on the MIMO channel systems based on Malaga turbulent channels and a Gamma-Gamma (GG) turbulence model. Authors in [32] and [33] presented a DWDM-FSO communication system that uses MIMO techniques with eight channels, each transmitting 2.5 Gbps

at a distance of 1.5 Km. The diversity combining techniques such as maximum-ratio combining (MRC), selection combining (SC), and equal-gain combining (EGC), the performance metrics evaluated comprise BER, outage probability (OP), and SNR.

2.1. Limitations of implementing FSO communication

Present RF wireless communication networks struggle to achieve data throughput coverage up to the last mile in 6G-IoT technology.

- High data rates and secure connections by FSO communications can resolve the challenge. The related work listed much research on FSO for 6G applications and the research issues with implementation and signal attenuations due to variations in atmospheric conditions.
- However, the signal loss due to the mechanical strain and temperature variations can be resolved if FSO is to reach its full potential for IoT deployments.
- Due to the interaction challenges between the FSO channel and FBG sensor, modulation affects system efficiency with FBG sensors. FSO links are susceptible to atmospheric turbulence, making modulation important for system robustness.

3. System design

This section contains thorough information about the proposed system design. Essentially, the suggested system begins with the transmitter, which includes a PRBS generator for test data, NRZ encoding for signal modulation, a 1552 nm CW laser for optical carrier generation, and a Mach-Zehnder Modulator (MZM) intensity modulation. The MIMO-based FSO channel improves dependability while reducing atmospheric turbulence effects such as scattering and absorption. Embedded FBG sensors inspect strain and temperature variations, allowing for real-time identification of position faults. A homogeneous FBG compensator rectifies wavelength shifts and chromatic dispersion while maintaining signal quality. The receiver evaluates performance measures involving BER and received signal power in different meteorological conditions. Environmental data gathered by FBG sensors is sent via MQTT to a web dashboard, offering real-time system insights and allowing for proactive modifications, making it perfect for secure, dependable communication and monitoring in harsh situations. The system parameters are listed in Table 1. Table 2.

3.1. Transmitter

As shown in Fig. 1, the transmitter section consists of a pseudo-random bit sequence generator (PRBS) set to 10 Gbps. The information is coded by non-return to zero-line codes, which connects the information signal to the intensity modulator. The carrier signal is generated by a continuous wave (CW) laser that is set to 1552 nm wavelength. The information signal in electrical form and the light signal from the CW laser are the inputs to the intensity modulator. The MZM has been used as an intensity modulator with two identical waveguides linked to two 3 dB couplers. The electro-optic effect in MZM allows for controlling refractive indices in waveguide branches by applying an externally given voltage. The output exhibits both constructive and destructive interference depending on the applied voltage. Amplitude modulation is simple and effective for intensity-based encoding of fibre Bragg grating sensor data, but turbulence causes amplitude fluctuations.

3.2. Free space optical channel

Mathematical models of turbulence models of FSO channel are described as Log-Normal and Gamma-Gamma for weak and moderate to strong levels. The log-normal model makes the assumption that changes in received intensity result from small random variations in the refrac-

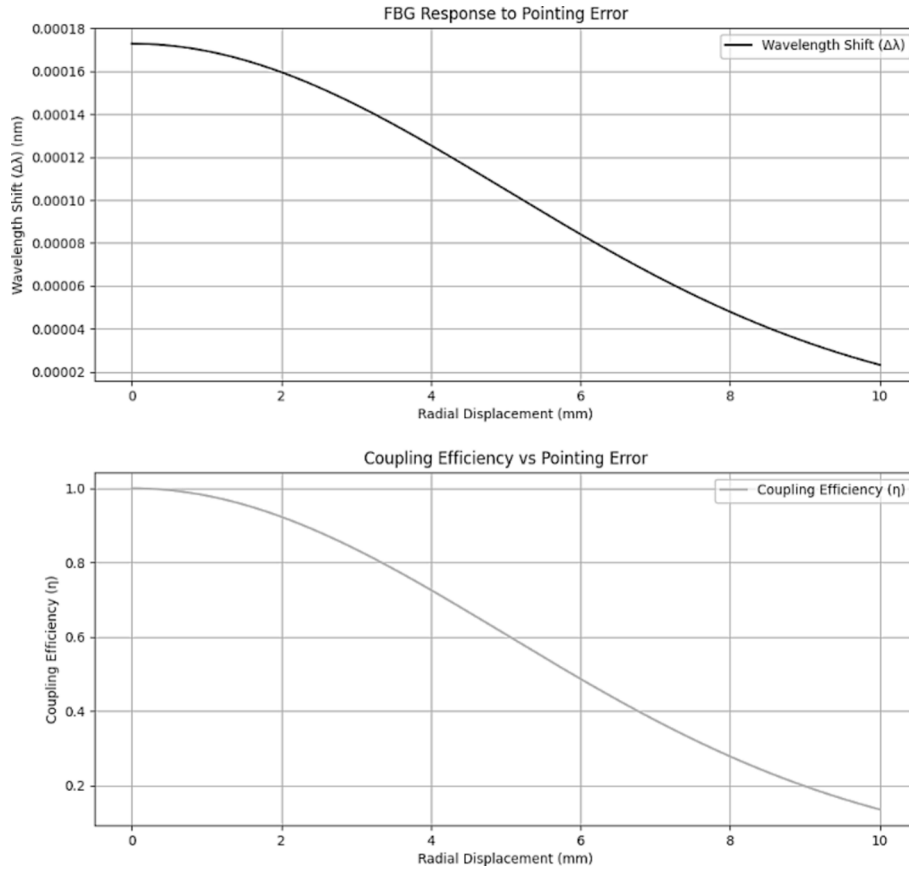


Fig. 6. Fiber Bragg Grating Response to Pointing Error.

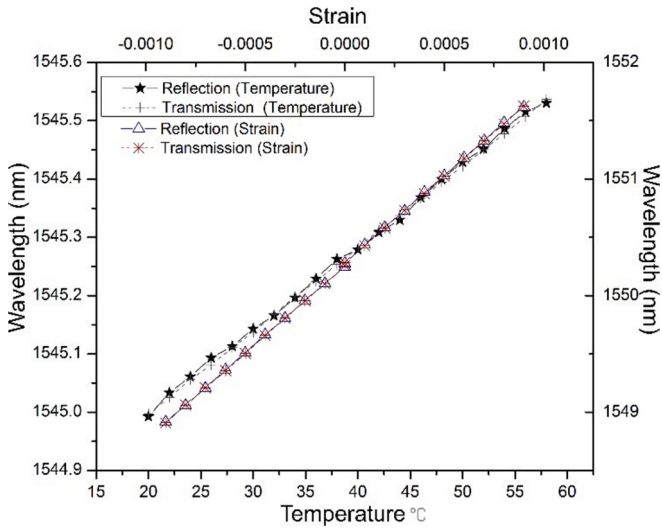


Fig. 7. Result of FBG sensor – wavelength shift due to temperature variation (Center wavelength set to 1545 nm) and due to applied strain (Center wavelength set to 1550 nm).

tive index. The probability density function (PDF) of intensity in Log-normal is shown as ref,

$$f_I(I) = \frac{1}{I\sqrt{2\pi\sigma_X^2}} \exp\left(-\frac{(\ln I - \mu_X)^2}{2\sigma_X^2}\right) \quad (1)$$

$$\sigma_X^2 = \ln(1 + \sigma_R^2)$$

Rytov variance :

$$\sigma_R^2 = 1.23 C_n^2 k^{7/6} L^{11/6}$$

The bit error rate for the on-off keying technique is derived as follows:

$$BER = \frac{1}{2} \operatorname{erfc}\left(\frac{\sqrt{P_r}}{\sqrt{N_0}}\right)$$

Log-normal is not valid for strong turbulence. The Gamma-Gamma model is appropriate for moderate-to-strong turbulence. It encompasses both small-scale and large-scale turbulence effects. The scintillation phenomenon is most accurately described by the Gamma-Gamma model, which results from atmospheric turbulence-induced random variation in the optical power received.

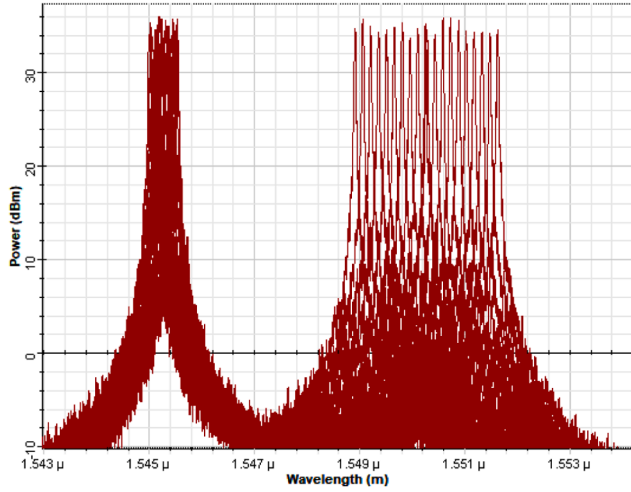
The PDF of Gamma-Gamma distribution is expressed as [34,35]:

$$f_I(I_{mn}) = \frac{2(\alpha\beta)^{\frac{\alpha+\beta}{2}} I_{mn}^{\frac{\alpha+\beta}{2}-1} K_{\alpha-\beta}(2\sqrt{\alpha\beta I_{mn}})}{\Gamma(\alpha)\Gamma(\beta)} \quad (2)$$

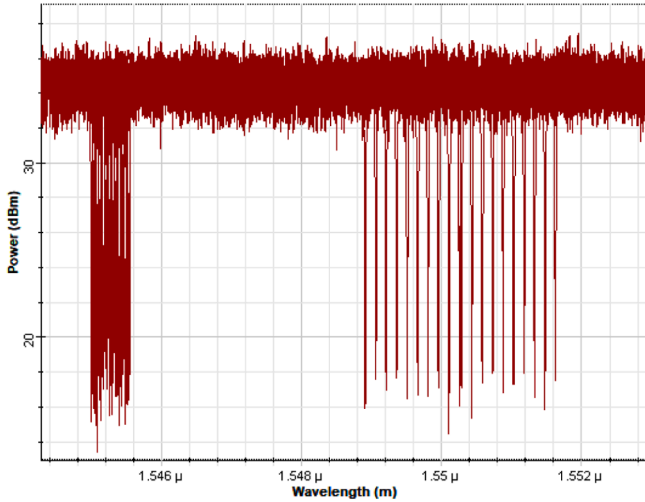
Where $\Gamma(*)$ – γ function, $K_v(*)$ – fifth order modified Bessel function of the II order type.

The most critical parameters α and β are small-scale and large-scale turbulence region parameters derived from atmospheric conditions using the following formula [35,36]

$$\alpha = \left[\exp\left(\frac{0.49\chi^2}{(1 + 0.18d^2 + 0.56\chi^{\frac{12}{5}})^{\frac{7}{6}}}\right) - 1 \right]^{-1} \quad (3)$$



(a)



(b)

Fig. 8. FBG sensor signal (a) Shifted and reflected signal powers, (b) Shifted and transmitted signal power.

$$\beta = \left[\exp \left(\frac{0.51\chi^2(1 + 0.69\chi^{\frac{12}{5}})^{\frac{-5}{6}}}{(1 + 0.9d^2 + 0.62d^2\chi^{\frac{12}{5}})^{\frac{7}{6}}} \right) - 1 \right]^{-1} \quad (4)$$

Where $\chi^2 = 0.5 C_n^2 k^{7/6} L^{11/6}$, $d = ((kD)^2/4L)(1/2)$, $k = 2\pi/\lambda$. D – Diameter of the receiver lens aperture, L – link distance in meters, depending on the degree of turbulence C_n^2 – The index of refractive structure parameter varies from $10^{-13} \text{m}^{-2/3}$ to $10^{-17} \text{m}^{-2/3}$ as strong to weak turbulence.

Bit error rate for on-off keying technique is derived as follows:

$$BER = \frac{1}{2} \int_0^\infty \text{erfc} \left(\sqrt{\frac{I}{N_0}} \right) p_I(I) dI$$

Both models are compared in this work, and the Gamma-Gamma model is used in FSO communication systems due to its versatility. In contrast, the Log-Normal model is better for mild turbulence. The observed BER for both scintillation models is discussed in the next section. In this work, the PDF of a Gamma-Gamma model is employed to model the effects of atmospheric turbulence. Also, The MIMO technique, which transmits and receives data using multiple antennas, is used in this system.

3.3. Estimation of signal attenuation due to weather conditions

A model for atmospheric attenuation was developed by [37],

$$\sigma(\text{dB/km}) = \frac{3.912}{V(\text{km})} \left(\frac{\lambda}{550\text{nm}} \right)^{-q(v)} \quad (5)$$

Where, σ – attenuation coefficient, V – Visibility, λ – Wavelength in nanometres. $q(v)$ – the particle size distribution coefficient that is described using the Kruse model and which is widely used.

$$q(v) = \begin{cases} 1.6V & \text{if } V > 50 \text{ km} \\ 1.3 & \text{if } 6 \text{ km} < V < 50 \text{ km} \\ 0.16V + 0.34 & \text{if } 1 \text{ km} < V < 6 \text{ km} \\ V - 0.5 & \text{if } 0.5 \text{ km} < V < 1 \text{ km} \\ 0 & \text{if } V < 0.5 \text{ km} \end{cases} \quad (6)$$

The revised model for estimating a loss in signal intensity resulting from fog and smoke is as follows [37,38]:

$$\sigma(\text{dB/km}) = \frac{17}{V} \left(\frac{\lambda}{550} \right)^{-q(v)} \quad (7)$$

Equation (6) evaluates the system to determine the atmospheric attenuation values in dB/km listed in Table II. These variables are used to evaluate the proposed system's performance in various visibility conditions.

The received optical power from the free space channel can be mathematically described by Vishwakarma and Swaminathan [16],

$$P_r = P_t \frac{D_r^2}{[D_t + (\theta^* Z)]^2} * 10^{\left(\frac{-\alpha Z}{10}\right)} \quad (8)$$

P_r – Received optical power, P_t – Transmitted power, D_r – Receiving antenna diameter, D_t – Transmitting antenna diameter, θ – Beam divergence angle, α – Specific attenuation Z – Transmission range.

Section IV analyses the proposed system's impact on power due to severe atmospheric attenuations, strain, and temperature effects and discusses the results.

3.4. Receiver

FBG sensors integrated at the receiver are used in the proposed system to monitor open environmental conditions that could affect the optical link's performance. A PIN photo-detector detects the optical signal at the receiver, and a low-pass Bessel filter filters out high-frequency components and noise. A current is produced by the photo-diode in direct proportion to the intensity of the incident light when exposed to light. After processing this electrical signal, the transmitted data is extracted, and the BER analysers examine the lowest BER performance.

3.5. Fiber Bragg grating sensor and the compensator

Temperature and strain will influence the wavelength and transfer functions in the Bragg grating of FBG sensors. Temperature changes and stress/strain variations affect the optical characteristics of an FBG. The fundamental equation determines the Bragg grating's wavelength,

$$\lambda_B = 2n_e \Lambda \quad (9)$$

Where λ_B is the Bragg wavelength, n_e is the effective index, and Λ is the grating periodicity.

The transmission and reflection depend upon the strength of the coupling and the phase mismatch. The reflection and transmission of the fibre grating sensor as a function of wavelength (λ) can be written as [39]

$$r(\lambda) = \frac{iK \sinh(QL)}{Q \cosh(QL) - i\Delta\beta \sinh(QL)} \quad (10)$$

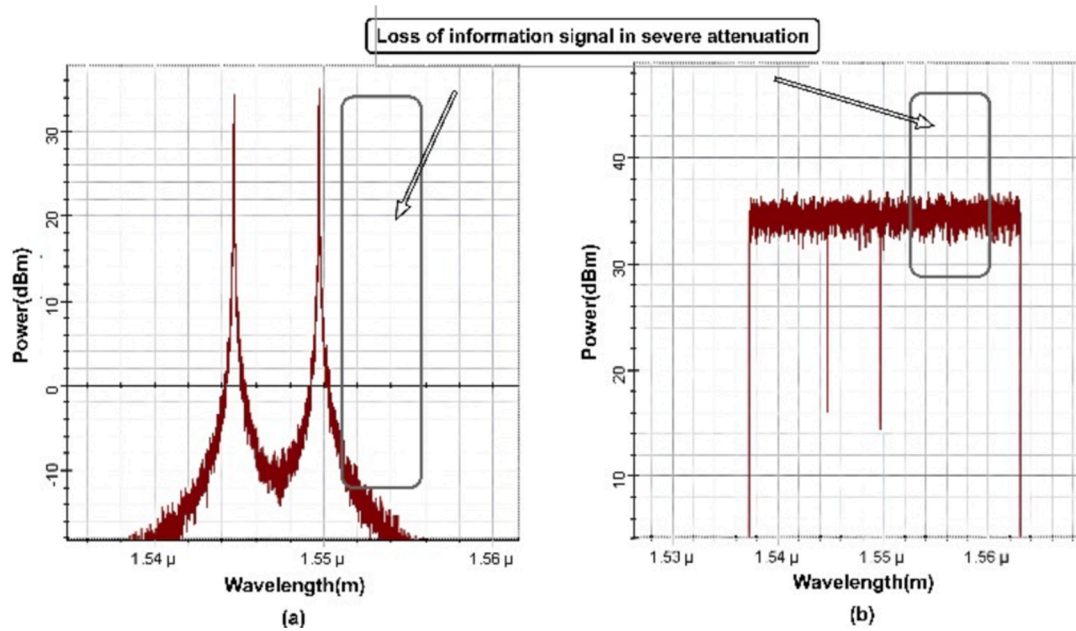


Fig. 9. FBG sensor signal in proposed system under severe attenuation by temperature and strain (a) Reflected signal power, (b) Transmitted signal power.

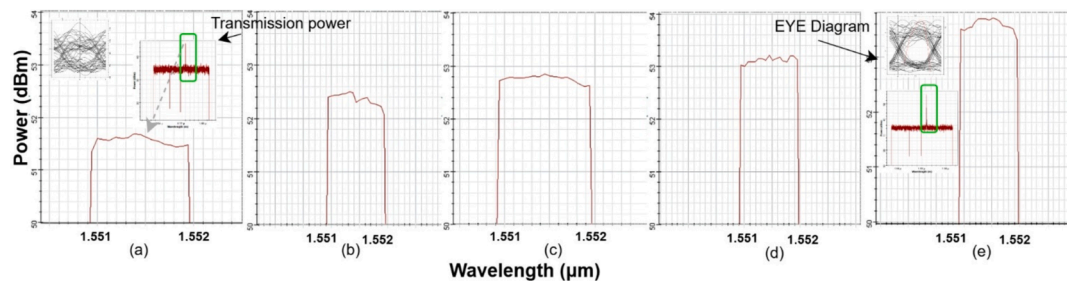


Fig. 10. Eye Diagrams – MIMO FSO and integrated FBG sensor system under clear air condition.

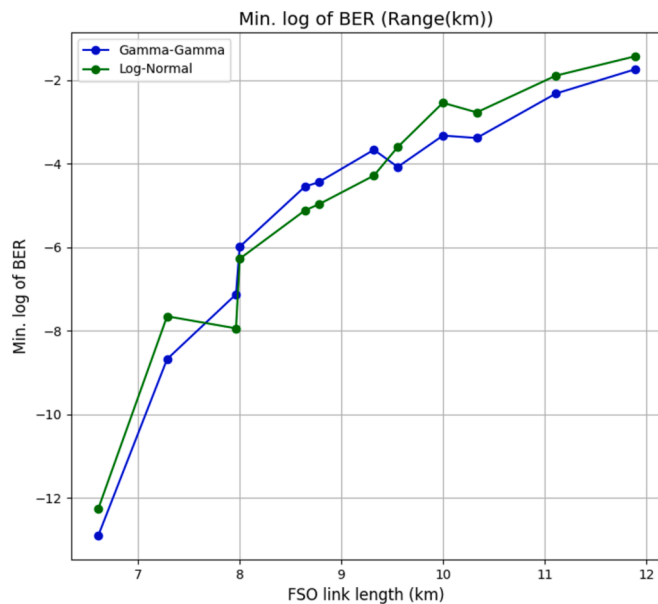


Fig. 11. BER vs Transmission link length for scintillation models Gamma-Gamma and Log-Normal.

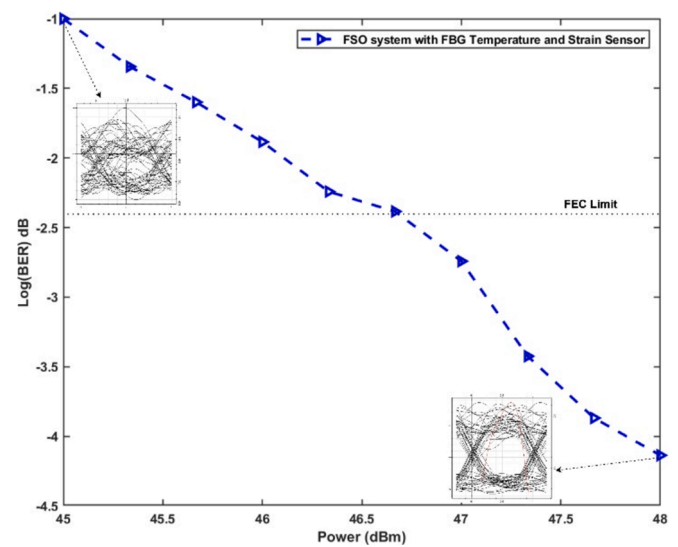


Fig. 12. BER vs Transmission power in a proposed system under clear atmospheric conditions and 10 km FSO link range with temperature and strain effects.

Table 3

Comparison of MIMO scheme, compensator with single FSO system in present work.

FSO system with Temperature and Strain effects	Log(BER) in dB	Received power in dB
FSO system	-3.12494	52.4
2 x 2 MIMO	-8.1691	55.6
2 x 2 MIMO with Compensator	-19.283	55.7
4 x 4 MIMO	-24.987	58.7
4 x 4 MIMO with Compensator	-35.018	58.9

Table 4

Comparison of various techniques used to achieve data rate and FSO link length.

Reference	Technique used	Channel model	Data rate	FSO link range
[41]	All-optical regenerate and forward	Gamma-Gamma	6 Gbps	1 km
[42]	Optical phase conjugation compensation	Experimental – Collimator (ThorlabsF810FC-1550)	5 Gbps	1 km
[21]	SAC-OCMA-based FSO system with DDDW code	Gamma-Gamma	1 Gbps	6.9 km
Proposed System	MIMO FSO- FBG sensor integration	Gamma-Gamma	10 Gbps	10 km

$$t(\lambda) = \frac{Q}{Q \cosh(QL) - i \Delta \beta \sinh(QL)} \quad (11)$$

Where

$$Q(\lambda) = \sqrt{|K|^2 - |\Delta \beta|^2}$$

$K(\lambda) = \Delta n \pi / \lambda$ is the coupling strength, Δn is the index modulation, L is the grating length, $\Delta \beta = 2\pi n_e (\lambda^{-1} - \lambda_B^{-1})$ is the phase mismatch.

• The sensing effect

The shift in the Bragg wavelength λ_B caused by external forces acting on the FBG is the source of the sensing effect. When λ_B changes, the spectra of reflection and transmission will vary as well. The equation is defined in terms of the external effects, which are assumed to be the longitudinal strain and temperature acting on the grating $\lambda_B = 2n_e \Lambda$ becomes [39]

$$\Delta \lambda_B = 2 \left(\frac{\partial n_e}{\partial T} \Delta T + n_e \frac{\partial \Lambda}{\partial T} \Delta T + \frac{\partial n_e}{\partial l} \Delta l + n_e \frac{\partial \Lambda}{\partial l} \Delta l \right) \quad (12)$$

which is rewritten as,

$$\frac{\Delta \lambda_{B1}}{\lambda_{B1}} = (\alpha_{TE} + \alpha_{TO}) \Delta T \quad (13)$$

$$\frac{\Delta \lambda_{B2}}{\lambda_{B2}} = (1 - p_e) \epsilon + (\alpha_{TE} + \alpha_{TO}) \Delta T \quad (14)$$

Where ϵ – longitudinal strain, ΔT – temperature change, p_e – photo-elastic coefficient, α_{TE} – thermal expansion coefficient, and α_{TO} – thermos optic coefficient.

This work uses two FBG sensors to sense temperature and the strain effects. Equations (12) and (13) are solved to get both the temperature and strain. The depicted arrangement is seen in Fig. 7.

• Pointing error sensing using FBG sensor

FBG sensors connected with the FSO channel detect the pointing

error, which involves mechanical and temperature changes brought on by misaligned beams at the receiver. These variations provide indirect information regarding pointing errors. λ_B given in Equation (9) specifies the Bragg wavelength and variations in Bragg wavelength by the strain can be represented by,

$$\Delta \lambda_B = \lambda_B ((1 - P_e) \epsilon - \zeta \Delta T) \quad (15)$$

The impact of pointing error, which is malalignment in beams, leads to temperature and strain effects in the FBG sensor.

• Compensator

The compensator connected to the receiver section in the system mitigates the effects of dispersion. Dispersion causes signal distortion and limits the achievable data rates. A dispersion compensating fibre (DCF) component is used, and properties such as length and dispersion coefficient are configured to overcome the impact of signal dispersion.

3.6. Mathematical modelling – FBG strain effects to pointing displacement

Considering r to be the pointing displacement, the induced strain ϵ on the receiver surface is modelled as,

$$\epsilon = \frac{F}{EA} \quad (16)$$

Where F is the force by pointing error, E is Young's modulus, and A is the cross-sectional area. Then $\Delta \lambda_B$ is by the effect of ϵ can be given as

$$\Delta \lambda_B = \lambda_B ((1 - P_e) \epsilon$$

Also, the misalignment impacts the coupling efficiency. η as,

$$\eta = \exp\left(-\frac{r^2}{2\omega_z^2}\right) \quad (17)$$

Where, ω_z – Beam waist.

Algorithm 1 discusses how the proposed system analyses signal attenuations and how interfacing the FSO link with the FBG sensor solves challenges. The wavelength shift of the FBG sensor was used to monitor strain and temperature, and FSO communication signal attenuation under various environments was examined.

Input: PRBS, Weather Conditions (C), FSO Link Range (R)

Laser Parameters: Wavelength=1552 nm, Transmission power=20 dBm, Laser linewidth=0.1MHz

Output: Bits Error Rates BER, Maximum Distance of FSO link Range (at FEC limits)

Integrated FSO communication FBG sensing system

Step 1: analyze_fso_system():

```

w_c = ["clear air"; "Very Light mist (Light rain)"; "Very Light Fog(Heavy rain)";
"Light Fog (Storm)"; "Moderate Fog"];
1 → Clear Air; 2 → Very Light mist (Light rain);
3 → Very Light Fog(Heavy rain);
4 → Light fog (Storm); 5 → Moderate Fog ∀ {1,2,3,4,5 ∈ w_c}
V → Visibility in km; σ → Attenuation

if (CC == 1)
    V = 20; σ = 0.54
else if (CC == 2)
    V = 5.9; σ = 2
else if (CC == 3)
    V = 1.9; σ = 6.9
else if (CC == 4)
    V = 0.77; σ = 18.3
else if (CC == 5)
    V = 0.5; σ = 28.9
End
for CC = 1: w_c
    Measure log (BER)
    Record the maximum supported FSO link
End

```

Step 2: Integrate FBG-IoT sensor ():

```

    Measure log (BER) and power in dB

```

Step 3: Compensation ():

```

    Measure log (BER) and power in dB

```

End

Calculate the Maximum distance of the FSO Link Range Measured at FEC (limits) and signal power required.

End

4. Results and discussions

The proposed FSO system integrated with the FBG IoT sensor has been analysed for received bit error rate (BER) and signal power, as well as an eye diagram. The sensor is tested with transmitted and reflected optical power. First, the FSO system is analysed under various atmospheric attenuation ranges of 0.54 dB/km, 2 dB/km, 6.9 dB/km, 18.3 dB/km, and 28.9 dB/km for clear air, very light mist (heavy rain), very light fog, light fog, and moderate fog, respectively.

Fig. 2 outlines the calculations of the log-normal and gamma-gamma turbulence models, which mostly concentrate on the PDF (probability density function) of the intensity variations generated by air turbulence. Bit Error Rate (BER) for modulation technique On-Off Keying (OOK) is computed using these PDF and it could be noted that the Gamma-Gamma model accommodates greater intensity withering under intense turbulence; however, the Log-Normal model has a narrower and less slanted distribution. Fig. 3 shows the maximum supported FSO link ranges under clear atmospheric conditions. Fig. 3(a) shows an eye diagram at 9.6 km with the wider eye open and the eye closing as the distance increased. The maximum supported FSO link range of 69.6 km under ideal conditions is shown in Fig. 3(c), which is in acceptable BER ($\leq 2 \times 10^{-3}$, i.e., FEC limit) [40].

Fig. 4 shows the BER received and the maximum FSO link range obtained under different weather conditions. In Fig. 4(a), the graph illustrates that an FSO channel's maximum distance is 69.6 km when attenuation is 0.54 dB/km, and in Fig. 4(b), the graph illustrates 23.4 km for 2 dB/km, 8.1 km for 6.9 dB/km, 3.2 km for 18.3 dB/km, and 2.4 km for 28.9 dB/km. The BER increases as the free space channel attenuation increases.

Fig. 5 displays the received power during the signal propagation through clear air, rain, and fog conditions. Various atmospheric conditions affect the received optical power. Fig. 5(a) shows reduced received optical power from -62.8 dBm to -64.9 dBm under a clear atmosphere. Fig. 5(b) shows the power results from atmospheric attenuations and FSO link length under rain and fog conditions.

Mist consists of tiny water droplets suspended in the air. Furthermore, raindrops in the atmosphere scatter and absorb the optical signal, significantly decreasing received optical power. The attenuation of the optical signal in light fog is moderate, leading to a moderate decrease in received optical power. Moderate fog contains a higher density of water droplets than light fog, leading to increased power reduction.

Previous results show that the signal attenuations in FSO systems are only due to atmospheric conditions. The FBG sensors analyse the impact of strain and temperature, which are shown in upcoming results and discussions. The FSO system under clear atmospheric conditions with temperature and strain effects has been considered for further investigation. The proposed system includes two FBG sensors, as shown in Fig. 1. The sensors sense the changes in temperature and strain, which will shift a Bragg wavelength. An effective analysis of the FBG sensor involves systematically varying the temperature and stress/strain levels. The wavelengths are set to 1545 nm and 1550 nm in the first and second FBG sensors, respectively.

Fig. 6 shows the FBG response and the coupling efficiency for pointing errors. λ_B was set to 1550 nm and E represents the strain, ω_z characterizes the Gaussian beam at the receiver side. The pointing displacement range r varies from 0 mm to 10 mm. Fig. 6a displays the pointing error detection as wavelength shift when displacement increased, and 6b shows a decrement in the coupling efficiency. It has been observed from the result that the FBG sensors technique provides quantifying minor displacements resulting from wavelength variations, and the sensitivity depends upon the strain and material characteristics. Pointing errors reduce coupling efficiency, resulting in an increased BER and decreased received power.

Fig. 7 displays the sensor's sensing capability as wavelengths shift. Equation (14) shows the variation in the Bragg wavelength when both strain and temperature are influenced. The FBG sensor is connected to

an interrogator, and the data acquisition module collects the sensor data. The wavelength shift information is subsequently transmitted to an IoT web server for real-time monitoring via the MQTT protocol.

Fig. 8 displays the resultant reflection and transmission spectra observed by the interrogation setup. This approach has created a setup in which the sensing system is interrogated at specific intervals. At intervals of one second, these interrogation points responses are monitored. Each second, there is a 2 °C increase in temperature and a 10e-5 range of strain increase. An interrogator emits a stream of optical signals every second, traversing the system and returning, followed by another stream transmitted a second later. Thus, with each time step, the system is interrogated.

A carrier wave signal wavelength in continuous wave laser is set at 1552 nm, and transmission power is 20 dBm, which carries 10 Gb/s data through the FSO channel in this system. Fig. 9 depicts the impact of severe attenuations due to temperature and strain in the FSO communication system. The information signal is severely lost, as shown by signal spectra from the FBG sensor component. The impact of temperature and the signal attenuation required to increase the transmission power upto 65 dBm and strain leads to pointing errors, which completely lose the information signal at the receiver.

The proposed system integrates MIMO FSO channels with the FBG sensors. A MIMO technique integrates signals from several apertures to expand the communication range. The system is analysed with MIMO channels, increasing the transmission power. Additionally, the compensator mitigates the dispersion in optical signals, leading to enhanced signal quality. Fig. 10 presents the results of received power while increasing the transmission power from 46 dBm, 46.6 dBm, 47 dBm, 47.6 dBm and 48 dBm. The results indicate that the impact of increased transmission power helps retrieve the information signal. Fig. 12 illustrates the BER that achieved the FEC limit for 46.6 dBm transmission power.

Fig. 11 shows the observed result for both scintillation models. The FSO link length is set from 6 km to 12 km. The BER is relatively low and constant for short FSO links, which have a small amount of scintillation. As the FSO link rises, the scintillation variance also increases, which results in higher swipes in the received signal intensity and an increased BER. The log-normal model is not suitable for strong-intensity fading and implies mild turbulence, which results in a gradual BER variation.

In the Gamma-Gamma turbulence model, both scales of turbulence are taken into consideration because the BER is slightly higher than the log-normal model when it comes to short FSO links that experience moderate turbulence observed from an 8 km link distance. As the length of the FSO link rises, the intensity of the scintillation increases as well, which results in a more severe fading of the signal, as observed from 9.6 km of FSO link length. This causes a non-linear increase in the BER. The Gamma-Gamma model is more realistic than the log-normal model in terms of depicting intensity fading for long channels.

Improvements in signal power and BER are analysed step by step and compared as listed in Table 3. The final investigation evaluated the performance of the MIMO FSO and the integrated FBG sensor system set to 46.6 dBm transmission power using single 1 x 1, dual 2 x 2, and 4 x 4 FSO channels with the compensator to overcome the impact of temperature and strain effects. The BER value is greatly reduced as a difference of 31.9 dB from the FSO system with severe attenuations to the system with 4 x 4 MIMO FSO channels and the compensator. The received power increased to 6.5 dBm. Table 4 shows the comparison of the present work with previous works.

5. Conclusion

The proposed system addresses severe signal losses due to various atmospheric conditions, temperature, and strain in the FSO channel integrated with the FBG sensors and compensator. The system significantly enhances the ability to analyse and monitor open environment data. The system shows the capability of adapting to varying levels of

laser power, thereby resolving the reduction in the intensity of the transmitted signal. MIMO channels improve system performance metrics, and embedding the compensator reduces dispersion and increases signal quality. Increased input CW laser power and 4 x 4 MIMO FSO channels with the compensator significantly improve BER and received power. The required transmission power has been reduced from 65 dBm to 46.6 dBm and can achieve the acceptable FSO link range of 10 km. The proposed system offers high-speed and secure communication, along with precise environmental measurements, facilitating the development of 6G wireless sensor IoT networks. The work can be extended by incorporating advanced modulation and signal processing techniques to support increased FSO link length and equalization of the required information.

CRediT authorship contribution statement

R. Arunachalam: Conceptualization, Data curation, Formal analysis, Investigation, Writing – original draft. **Rupali Singh:** Writing – review & editing, Supervision, Project administration, Formal analysis. **M. Vinoth Kumar:** Methodology, Resources, Software, Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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